

Ayaan Hafeez

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SUMMARY

Computer Engineering student with production experience spanning AI-powered backend systems, full-stack development, and low-level hardware programming. Shipped LLM pipelines, NL2SQL chatbots, and REST APIs in Python and FastAPI at Elev8AI, and built projects ranging from an Android face-recognition system in Kotlin to a bare-metal OS kernel in C and ARM Assembly.

EDUCATION

University of Alberta

Edmonton, AB

BSc in Computer Engineering Co-op Software Option

May 2028

- **Related Coursework:** Data Structures & Algorithms, Object Oriented Programming, Computer Organization & Assembly Language, Software Requirements & SDLC, Digital Logic Design, Digital Signals

TECHNICAL SKILLS

Languages: Python, Kotlin, Java, C/C++, JavaScript, SQL, NoSQL, Bash, VHDL, Assembly (ARM, MIPS)

Frameworks/Libraries: FastAPI, Flask, Node.js, React, LangChain, SQLAlchemy, scikit-learn, NumPy, pandas

Developer Tools: Git, Docker, PostgreSQL, Redis, ChromaDB, Selenium, Stripe API, AWS (Textract, Rekognition)

Hardware: FPGA (Xilinx Zybo), Arduino, ARM Cortex M4, Raspberry Pi, Analog Discovery, BioAMP EXG

EXPERIENCE

AI Developer

May 2025 – Present

Elev8AI

Remote

- Architected an NL2SQL network-monitoring chatbot using LangChain, Ollama, ChromaDB, and PostgreSQL with RAG retrieval and session-aware context management, achieving 85% answer accuracy on internal benchmarks
- Engineered multi-turn LangChain SQL agents and prompt pipelines for an AI trip-planning platform using OpenAI GPT-4, enabling structured itinerary generation across database-backed and general-knowledge destinations
- Built the backend of a social media platform with JWT auth, Stripe payment processing, and AWS Textract/Rekognition ID verification, and a Selenium web scraper that extracted and normalized 100+ REST endpoints from JS-rendered documentation into OpenAPI-compatible schemas reducing developer hours by 90%
- Delivered an Android face-recognition attendance system using Kotlin, Jetpack Compose, CameraX, and Google ML Kit with real-time face detection, head-pose estimation, and frame-level deduplication

PROJECTS

Bare Metal Raspberry Pi OS | C, ARM Assembly, Raspberry Pi

Apr 2026

- Built a bare-metal OS kernel for Raspberry Pi 3 in C and ARM Assembly across 6 subsystems: bootloader, UART/GPIO drivers, interrupt controller, preemptive scheduler, system calls, and MMU-backed virtual memory, with zero external library or OS dependencies
- Implemented a preemptive process scheduler and fork-based process isolation using ARM exception levels and a per-process page table, supporting concurrent execution across all 4 CPU cores

FocusNode – Browser Extension | React, Vite, Chrome Extensions API, JavaScript

Jan 2026

- Built a Chrome productivity extension using React and Vite that intercepts and redirects user-defined domains during focus sessions, achieving tab blocking within 50 ms of navigation
- Architected persistent state management using Chrome Storage APIs to synchronize blocklists and focus state across browser restarts, maintaining 100% state retention

Tuesday.com – Project Management App | Node.js, Express.js, MongoDB, JavaScript

Feb 2025

- Led a 5-person team to design and ship a full-stack project management application using Node.js, Express.js, and MongoDB with AI-driven task automation, delivering a functional product within a 48-hour hackathon window
- Integrated an AI task suggestion engine that achieved a 60% reduction in decision-making time during simulated workflows, validated through structured user testing

EMG Controller | Arduino, BioAmp EXG Pills, Python, C

Nov 2024

- Built a real-time EMG-based input controller using BioAmp EXG Pills, Arduino, and Python/C signal processing that converted muscle contractions to keystrokes with <50 ms end-to-end latency and 40% fewer false activations